The D_IV^5 Ribbon: Fermion Quantum Numbers as the Holonomy of a Spinor Winding

A geometric framework in which every fermion quantum number is one feature of a single framed, twisted, wound ribbon assembled on the bounded symmetric domain D_IV^5.

Author: Casey Koons CI co-authors: Keeper, Lyra, Elie, Grace Status: Framework draft v0.5 — 2026-07-12. *Honestly tiered.* This paper presents a geometric realization, not a completed derivation. Section 7 is a complete ledger of what is derived, what is a candidate mechanism, and what remains open. Nothing here is claimed as proved that is not proved in Section 7.

> ★ CORRECTION STAMP (2026-08-16, K1624 — Koide section SUPERSEDED, do not cite as live). The v0.4 changelog item 1 below ("Koide = rank/N_c → CONDITIONAL FORCED; A^2=rank from the Bergman overlap") is RETIRED. The Koide FORCING was ruled BREAK on 2026-08-06 (K1221): the addresses {5/2,3/2,0} can never be B_2-symmetric (no Weyl/isotropy mechanism), and the c-function magnitude route was retracted (F669). Current tier: Koide = IDENTIFIED — the *relation* Q=2/3 stays banked (exact to 0.001%), the *derivation* does not; Koide is open in physics. Two specific corrections: (a) "A^2=rank" is RETIRED — it is a rank-2 coincidence (equal-norm reads A^2=2 at every rank); the honest statement is "A^2=2 from equal-norm," and the derived form is (1+A^2/2)/N_gen, NOT rank/N_c. (b) the lepton nested-tower / genus-{5,3,0} strata and $c_5/c_3=\Gamma(5)/\pi^2$ (used in the "muon closed via strata" reading) are RETRACTED (F669/K853/K928/K1221 — a named repeat offender; the ν-addresses {5/2,3/2,0} are banked, the genus-{5,3,0} sub-domains are NOT). The τ falsifier m_τ=1776.969 (0.91σ) is Koide's 1981 prediction, not BST's. See K1619/K1624 and Paper_The_Operation_One_Page_v0.1 for the current honest state.

v0.5 changelog (the fermion sector nears whole — full detail in **`Keeper_Mass_Sector_and_Falsifier_Reframe_Results_2026-07-11.md`** Sections 2e–2m): 1. Top α-partition (up-type boundary sector closes). m_t = (1−α)·v/$\sqrt{2}$ = 172.75 GeV (0.03%); with m_c = α·v/$\sqrt{2}$, m_t + m_c = v/$\sqrt{2}$ at 0.007% — the top and charm partition one boundary mode at α (top saturates minus one α-shell; the deficit IS the charm). Cross-checked v-independently by m_t/m_b = 42 = C_2·g. 2. Charged leptons: muon closed, tau exact and forward. The mass rides the Harish-Chandra formal degree d(ν) (not the bare FK residue). Muon = (24/π^2)^{C_2} = 206.76 (0.003%). Tau = 49·71 − $\sqrt{\pi}$ = 3477.2275 (0.0000%) — the boundary sum (bulk 343 + edge 3136) minus the cone-tip residue $\sqrt{\pi} = \Gamma(-3/2)/V(B^3) = (4\sqrt{\pi}/3)/(4/3)$, the $-\sqrt{\pi}$ now *forward* (Casey's "4/3 is a 3-ball volume"). The tau's integral is BST's first evaluated boundary integral (a build; structure-set). 3. A structural keystone (Section 4a): n_C = rank + N_c = "two crossings + three commitments." The tau's 3-ball is the a = n_C − rank = N_c = 3 transverse Peirce directions (T2511, F368, banked) — the three color/commitment channels. 4. Neutrino → Majorana (near-forced, K673). The chargeless "odd-one-out": its RH partner sits at ν=9/2 with a *negative* Harish-Chandra formal degree (d(5−ν)=−d(ν)) — non-unitary, strictly not a state. Doubly confirmed: the banked seesaw masses are intrinsically Majorana. Flips our sharpest falsifier: pred_004 from "0$\nu\beta\beta$ never" to "0$\nu\beta\beta$ at the 1–4 meV floor" (pending Cal co-sign + the explicit γ^5 intertwiner). Section 6 and Section 7 updated. 5. CKM forcing opened as the next frontier (identified, not yet forced; the load-bearing gap is forcing the three quark localizations {r_i}).

v0.4 changelog (the two afternoon closes — Koide derived, the absolute scale mapped): 1. Koide = rank/N_c moves from IDENTIFIED to CONDITIONAL FORCED (near-bank; K672). The relation collapses to one angle — the tilt θ of the $\sqrt{}$mass vector from the democratic axis — and the data sits at θ = 45.00° exactly. Rank-2 forces it: N_gen = rank+1 = 3 (F86); the colorless hierarchy fills the rank traceless directions democratically → A^2 = rank = 2 (confirmed by lepton data to 0.02%) → Q = 2/3. The derivation carries its own falsifier and passes: only colorless obeys (up-type Koide 0.849, down-type 0.731 — only leptons hit 2/3). One gate remains: derive A^2 = rank from the Bergman overlap at the F86 strata. 2. Absolute scale mapped (F511): one dimensionful input → m_e (0.03%) → m_p → v = m_p^2/(g·m_e) = 246.1 GeV (0.04%); 8 of 9 charged fermion masses fall out of m_e and v with no second per-sector anchor. Softest charged piece: the down absolute anchor m_d (~5%). 3. The up-interior doublet-flip hypothesis (v0.3 open lane) is CLOSED as disfavored — the up reads its masses through the boundary, not a flip of the down's {1,3,5}.

v0.3 changelog (the mass sector rebuilt — the v0.2 "structural miss" was a wrong-target error, now dissolved):

1. The mass sector is REBUILT from "confirmed structural miss" to "mechanized in two halves." The v0.2 miss compared the bare geometric mass to *constituent* quark masses; the correct target is the *current* (Lagrangian) mass (the commitment/boundary principle: geometry commits current, QCD glue dresses to constituent). Against current masses the odd degrees {1,3,5} are *right* — read through the Faraut–Koranyi generalized Pochhammer (ν)_λ at ν = N_c, not the naive $(1-r^2)$ ladder. See the rewritten Section 3.2. 2. DOWN quarks BANKED (F506): d:s:b = 1:20:840, with s/d = (N_c+1)(N_c+2) = 20 forced (0.5% vs current), zero parameters, target-innocent; single-row forced by the Q^5 cohomology ring $\mathbb{Z}[h]/h^6$ (Elie). 3. UP quarks — endpoints forced, two masses predicted: the up-type sits on the Shilov boundary; m_top = v/$\sqrt{2}$ = 174 GeV (0.78%) and m_charm = α·v/$\sqrt{2}$ = 1269 MeV (0.05%) predicted from (m_p, m_e, g, N_max) with zero heavy-quark input, where v = m_p^2/(g·m_e) is forced (F509). The gen-1 inversion (m_u < m_d) is *forced* (up = ground state n=0, the doublet-flip of the down; its ground sits below the down's). 4. CHARGED LEPTONS closed by Koide = rank/N_c (0.001%): with m_τ/m_e = 49·71 the muon follows at 0.05%, zero muon input. (Identified at v0.3; *derived from rank-2 at v0.4* — see the v0.4 changelog and Section 3.2.) 5. The mass sector is TWO mechanisms, not one: quark boundary/bulk ladders + lepton Koide. The membrane-refraction spec is retained as the physical picture of the up-type boundary emission, not as a rescue of a miss. 6. Carried from v0.2: charge grounded on proved T2470; the "one −1 does all three" unification retracted; CP's up-down structure sourced (Pin(2)/Möbius).

Abstract

We propose that the quantum numbers of a fermion — its type, generation, mass, electric charge, spin, and matter/antimatter character — are not six independent labels but six geometric features of a single object: a framed (twisted) ribbon winding assembled on the bounded symmetric domain D_IV^5 = SO$_0$(5,2)/[SO(5)×SO(2)], the substrate of Bubble Spacetime Theory (BST). The substrate emits discrete information; the holomorphic boundary — which we call the membrane — assembles a self-sustaining spinor oscillator from it and dresses it in energy. This is the discrete/continuous principle promoted from a reading rule to a manufacturing process: the shape is information (discrete, forced), the energy is dressing (continuous, curvature).

Five geometric features of the ribbon carry the five families of fermion quantum numbers:

feature	quantum number
shape (commitment count 1/2/3)	type: lepton / meson / baryon
pitch (excitation degree ℓ, radius)	generation and mass
in-out winding (N_c=3 wraps around rank-2 posts)	electric charge
$\mathbb{Z}_2$ half-twist (ribbon framing)	spin
reading-side (which side of the $\mathbb{Z}_2$)	matter / antimatter

We show, with explicit tiering, that: (i) electric charge (magnitude and sign) is the SO(2)-weight operator of D_IV^5 (proved, T2470), quantized in units of 1/N_c with exactly N_c+1 = 4 magnitudes {0, 1/3, 2/3, 1}, capped at 1, no exotic charges — passing the BST Five-Absence filter; (ii) spin-1/2 is the $\mathbb{Z}_2$ half-twist of the ribbon, native to D_IV^5 as a type-IV (spin-factor) domain, and carries Fermi statistics automatically by the ribbon spin-statistics identity; and (iii) CPT is exact by construction, because the antiparticle is the *same* oscillator read from the other side of the membrane's $\mathbb{Z}_2$. Spin, the charge sign, and matter/antimatter each carry a "−1," but they are three distinct operations — the spinor 2π holonomy, the SO(2)-weight sign, and charge conjugation — related by the geometry, not one operation (a v0.1 over-merge, corrected). The mass sector is mechanized in two halves (Section 3.2): the quarks by boundary/bulk ladders — down-type banked (d:s:b = 1:20:840, s/d = 20 forced, 0.5% vs current mass), up-type with the top and charm masses *predicted* (174 GeV at 0.78%, 1269 MeV at 0.05%) from forced quantities with zero heavy-quark input — and the charged leptons closed by the Koide relation Q = rank/N_c (0.001%), which we derive from rank-2 geometry: it is the statement that the $\sqrt{\text{mass}}$ vector tilts at exactly 45° from the democratic axis, forced by N_gen = rank+1 and a colorless filling amplitude A^2 = rank, with a self-carrying falsifier the quarks pass (near-forced; one gate open). The net picture (Section 3.2, F511): from a single dimensionful input, 8 of the 9 charged fermion masses follow from m_e and the forced electroweak scale v =

$m\_p^2/(g\cdot m\_e)$. The up ground value is soft at n=0, the up-type interior and the neutrinos remain open, and the down absolute anchor is the softest charged piece; all are named in Section 7.

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1. Introduction

1.1 The one mechanism

BST derives the Standard Model from one geometry, D_IV^5, and five forced integers (rank = 2, N_c = 3, n_C = 5, $C_2 = 6$, g = 7, with $N\_max = N\_c^3 \cdot n\_C + rank = 137$). The present paper isolates a single organizing claim about *fermions*:

> A particle is a standing wave — a self-sustaining oscillator — that the membrane assembles from committed information and releases. Every quantum number of the particle is a geometric feature of that wave. There is one mechanism, and the quantum numbers are its coordinates.

This is not "a theory of everything asserted up front." It is a geometric realization whose numerical content is delivered — or falsified — by an explicit construction on D_IV^5 (Section 8). We state throughout, and collect in Section 7, exactly which parts are proved, which are candidate mechanisms, and which are open.

1.2 What a reader should take from this paper

For the referee: a specific, falsifiable claim that fermion quantum numbers are winding-geometry, with a hard ledger of evidence tiers and a single construction that decides the open items.

For the bright high-schooler: *imagine a rubber band wound a few times around two pegs, then given a half-twist and dressed in energy. How many times it winds is its electric charge. The half-twist is its spin. Which way you read it — front or back — is whether it is matter or antimatter. Every "particle" is one such wound, twisted band, and all the confusing labels in a physics textbook are just different ways of describing the band's shape.*

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2. The membrane and the two steps

D_IV^5 is a bounded symmetric domain of rank 2 and complex dimension 5. Its Shilov boundary is $(S^4 \times S^1)/\mathbb{Z}_2$. We call this boundary the membrane: it is the holomorphic screen on which the interior (bulk) data is written, in the sense of the holographic structure intrinsic to SO(5,2) ⊃ SO(4,2).

A fermion is produced in two steps:

1. Emission of information (discrete). The substrate's commitment cycle forms a shape: a winding pattern, an excitation degree, a framing. This is a *count* — forced, integer-valued, target-innocent.
2. Dressing in energy (continuous). The membrane assembles a self-sustaining oscillator carrying that shape and dresses it in energy through the Bergman metric, which diverges toward the membrane. This is the *curvature* — continuous, scale-carrying.

These two steps are the discrete/continuous principle of BST, seen as manufacturing rather than as bookkeeping. The particle *is* the assembled oscillator (not something emitted by an oscillator): this is the quantum-field creation operator $a^\dagger|0\rangle$ given a mechanism. Because the oscillator is self-sustaining once released, matter is stable without an ongoing energy source; because it is assembled by one process from one template, all electrons are identical; because it is an oscillator, it has an intrinsic frequency, and $E = \hbar\omega$ identifies that frequency with its mass.

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3. The five features

3.1 Shape → type (lepton, meson, baryon)

The membrane emits a color-singlet before it can release a particle, and N_c = 3 sets how many commitments a singlet requires. Counting commitments builds an (n−1)-simplex:

- 1 commitment → a point (0-simplex): a lepton — pointlike, colorless, lightest.
- 2 commitments → a line (1-simplex): a meson — a flux tube between two endpoints.
- 3 commitments → a triangle (2-simplex): a baryon — heaviest, the confined triple.

Two consequences follow. First, confinement is the statement that the pump releases one pulse per completed singlet. A single colored commitment is never emitted alone; it waits for its partners. Confinement is not a separate force here — it is the emission granularity N_c = 3. Second, the per-commitment energy is m_p/N_c ≈ 313 MeV (the light-quark constituent floor), consistent with the proton being the three-commitment emission product.

Energy grows monotonically with the dimensional extent within a generation (electron 0.5 MeV < pion 140 MeV < proton 938 MeV): “mass = occupied volume,” stated as the simplex dimension. (Across generations the pitch ladder of Section 3.2 stacks on top, so a heavy lepton excitation can outweigh a light baryon.)

3.2 Pitch → generation and mass (MECHANIZED IN TWO HALVES — quark ladders and lepton Koide)

Generations are excitation levels. On the rank-2 domain the three natural strata (Korányi–Wolf: bulk / Cartan slice / Shilov boundary) give exactly three, and Elie’s blind, mass-independent assignment reads them off the odd cohomology of the compact dual Q^5:

$\ell \in \{1, 3, 5\}$ — the odd cohomology cycles $\{h^1, h^3, h^5\}$ (T1929, proved; forced count, not fitted, zero mass reference).

The mass sector turns on one target ruling, then splits into two mechanisms.

The target ruling (the commitment/boundary principle). The geometry commits the *current* (bare, Lagrangian) mass; the ~300 MeV QCD glue is a separate *additive* dressing to the constituent mass. So every geometric mass prediction must be read against the current mass, not the constituent. This is not a post-hoc choice: it is the discrete/continuous principle at the particle level (the commitment is the conserved skeleton; the glue is the sheddable balloon). *This ruling dissolves a v0.2 error:* the “confirmed structural miss” of v0.2 compared the bare mass to the *constituent* ratios (s/d ≈ 1.6), where the heavy glue crushes the pattern. Against current masses the odd degrees are correct — and the mechanism is not the naive $(1-r^2)$ ladder but the domain’s own hypergeometric structure.

Down-type quarks — BANKED (F506). Read the three forced degrees through the Faraut–Koranyi generalized Pochhammer $(\nu)\_\lambda$ at ν = N_c = 3 (the Wallach threshold k_min), the native object of the Bergman machinery. The down ladder $\{(3)_1, (3)_3, (3)_5\} = \{3, 60, 2520\}$ gives

d : s : b = 1 : 20 : 840, with s/d = (N_c+1)(N_c+2) = 20 forced — zero parameters, target-innocent. Against current masses: s/d = 20.0 (obs 19.9, 0.5%); b/d = 840 (obs ≈ 890, 6%). Single-row is *forced by topology*: the Q^5 cohomology ring is $\mathbb{Z}[h]/h^6$, rank-1 in every degree, so every generation class is a single-row K-type (k,0) — two-row classes are absent (Elie). This is a genuine bank: forced, in-corpus, zero-parameter.

Up-type quarks — endpoints forced, two masses predicted. The up is the excitation of the down: exciting the gen-1 down = the Pin(2)/Möbius doublet flip (T_3: −½ → +½), which lands it on the *up-type* ladder, localized toward the Shilov boundary (the higher SO(2)-weight is more boundary-concentrated, T2470; independently, an excitation sits at larger radius). The ladder’s endpoints are then forced with zero fit:

- top: the boundary mode itself, Yukawa y_t = 1 (Cauchy–Schwarz saturation of the fermion–Higgs overlap), so m_top = $v/\sqrt{2}$ = 174 GeV (obs 172.7, 0.78%), where the electroweak scale v = $m\_p^2/(g\cdot m\_e)$ = 246.1 GeV is forced with no top input (F509) — the heaviest known particle predicted from two light masses and an integer;
- charm: one boundary shell in, coupling with the boundary’s own coupling α = 1/N_max, so m_charm = $\alpha \cdot v/\sqrt{2}$ = 1269 MeV (obs 1270, 0.05%) — the charm Yukawa *is* the fine-structure constant;

- up: the ground state n = 0 (the doublet-flip lands here), which sits *below* the down ground (degree 1) — so the gen-1 inversion m_u < m_d is forced, not an anomaly (and it is why the proton is stable and matter exists). Its value is soft at n = 0 (the constant mode is the least-reliably-computed overlap; the naive value undershoots ~22%, a computation limit named in Section 7, not a structural gap).

The up-type interior rung (the charm's generation degree) is the open piece: the clean hypothesis is that the up-type ladder is the doublet-flip of the down's {1,3,5} cohomology (Section 7, Open).

Charged leptons — closed by Koide. The leptons are colorless, so they carry neither the down-type bulk ladder nor the up-type boundary coupling (both use N_c). Their sector closes on its own most-precise relation — the Koide relation, which sits exactly at a BST ratio:

$Q = (m\_e + m\_\mu + m\_\tau) / (\sqrt{m\_e} + \sqrt{m\_\mu} + \sqrt{m\_\tau})^2 = 2/3 = \text{rank}/N\_c$ (0.001%).

With one further forced form, $m\_\tau/m\_e = 49{\cdot}71 = g^2{\cdot}(2\text{^}C_2 + g)$ (T2003), the Koide constraint forces the muon at $m\_\mu/m\_e = 206.9$ (obs 206.77, 0.05%), with no muon input.

This is now derived from rank-2, not merely identified (near-forced, K672). Writing the three $\sqrt{}$masses as a democratic constant plus a hierarchy oscillation, $\sqrt{m\_k} = M(1 + A{\cdot}\cos(\delta + 2\pi k/3))$, the relation collapses to one geometric fact: $Q = (1 + A^2/2)/N\_gen = 1/(N\_gen{\cdot}\cos^2\theta)$, where θ is the tilt of the $\sqrt{}$mass vector away from the "all generations equal" axis. The data sits at $\theta = 45.00°$ exactly. Rank-2 forces that tilt: (i) N_gen = rank + 1 = 3 (F86 — the three generations are the rank+1 support strata of D_IV^5), so the $\sqrt{}$mass vector lives in 1 democratic + rank hierarchy directions; (ii) the colorless hierarchy fills those rank directions democratically, unit amplitude each, giving A^2 = rank = 2 — confirmed target-innocently by the lepton data to 0.02% (observed $A^2 = 1.9997$). Then Q = (2 + rank)/(2(rank + 1)) = 2/3. The derivation carries its own falsifier and passes: only a colorless sector fills the rank-2 space cleanly, so colored quarks must miss 2/3 — and they do (up-type Koide = 0.849, down-type = 0.731; only the leptons hit 2/3). "Colorless" is thus the *physical condition* for the clean filling, not a label. The one remaining gate is deriving A^2 = rank from the Bergman overlap norm at the three F86 strata (Section 7).

Conclusion: mass is mechanized in two halves — not one grand law. The quarks run on boundary/bulk ladders (down: FK bulk {1,3,5}; up: boundary overlap, y = exp(−distance to the boundary), one shell = α); the leptons on Koide = rank/N_c. Each is forced by its own BST structure; the honest picture is two mechanisms, and it is stronger for being two solid halves than one over-reaching law. The membrane-refraction spec (separate) is the physical picture of the up-type boundary emission, retained — not a rescue of a miss, because there is no longer a miss to rescue.

3.3 In-out winding → electric charge

The ribbon winds N_c = 3 times around two posts (Section 4 identifies the posts with the rank-2 torus of D_IV^5). Each winding wraps a post either "in-out" or "out-in." Electric charge counts the in-out windings:

|q| = (number of in-out windings) / N_c.

For n ∈ {0, 1, 2, 3} this gives magnitudes {0, 1/3, 2/3, 1} — exactly the four fermion charge magnitudes of the Standard Model, and three structural results follow, all target-innocent (they use N_c = 3, not the observed charges):

1. Charge is quantized in thirds *because there are three windings.*
2. There are exactly N_c + 1 = 4 magnitudes — matching both the four charges and the four fermion types per generation (ν, d, u, e).
3. No charge exceeds 1, and none is a non-third: a genuine forbidden-list prediction. This passes the BST Five-Absence filter — it reproduces the SM charges and forbids the exotic (e.g. GUT leptoquark) charges BST already forbids.

The grounding (v0.2, proved): electric charge — magnitude *and* sign — is the SO(2)-weight operator of D_IV^5 (T2470, proved). The magnitude is the weight's size (thirds, N_c+1 values); the sign is the weight-sign — which way the winding runs. The winding-count picture above is a faithful *reading* of that weight (a different, N_c-scaled count), not an independent source. In particular the $(-1)^n$ form of the sign (v0.1, old Section 4) is a *repackaging* *of the assignment*, not a mechanism — the real sign is T2470's weight-sign. So charge sits in Solid (Section 7) on a proved footing, not on a winding holonomy.

3.4 $\mathbb{Z}_2$ half-twist → spin

Give the winding width — make it a ribbon — and put in one half-twist (a Möbius). Trace it: one turn (2π) returns it flipped, −1; two turns (4π) return it to itself, +1. This is not a metaphor for spin-½ — it *is* spin-½: the double cover Spin ⊃ SO, the topological spin θ = e^(2πi s) with s = 1/2 giving $\theta = -1$, the Dirac belt trick. Integer twists give integer spin (bosons); half-twists give half-integer spin (fermions).

Two facts make this native rather than imposed:

- Spin-statistics is free. In a ribbon (framed) category, twisting a particle by 2π equals exchanging two of them; the half-twist that gives spin-½ therefore gives the −1 under exchange — Fermi statistics, Pauli exclusion — with no additional postulate.
- The spinor is native to D_IV^5. Type-IV bounded symmetric domains are the spin-factor Jordan/Clifford domains; the spinor is the algebraic backbone of the domain, not an import. A fermion is the substrate-Dirac field — the square root of the oscillator ($\gamma\cdot\partial$ squares to ∂^2) — and the half-twist is that square-root operation. Every half-integer that recurs in BST (the ρ-vector (5/2, 3/2), the $\sqrt{}$ in the Cabibbo angle, "n_C is odd so you get a $\sqrt{}$") is the spinor structure surfacing.

3.5 Reading-side → matter and antimatter

The membrane's $\mathbb{Z}_2$ has two sides. The same assembled oscillator read from the wrong side is the antiparticle — the Feynman–Stückelberg reading (an antiparticle is a particle read backward in time), here given a mechanism (the two sides of the membrane are the two time directions).

Three facts collapse into one statement:

- CPT is exact — because the antiparticle is not an independent object that happens to match; it is the same oscillator read from the other side, so it must be the exact mirror. This is why CPT is the one symmetry never violated: it is not a symmetry between two things but one thing read two ways.
- CP is violated — the membrane is not a perfect mirror; the 45° cant tilts it, so one reading is preferred.
- Matter dominates — that preference accumulates into the baryon asymmetry. (Its magnitude, η_B, is a separate continuous quantity; only its *existence* and *sign* are the cant here — see Section 7.)

Annihilation is clean in this language: bringing matter and antimatter together reads the same oscillator from both sides at once; it superposes with its own reverse, cancels, and the membrane reclaims the energy.

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4. Three distinct −1's — spin, charge-sign, matter/antimatter (v0.1 over-merge, corrected)

v0.1 claimed these three were one operation — "the spinor $\sqrt{}$ appears three times, the same −1 each time." That was an over-merge, and the decider-day audit (Grace, via proved T2470) corrected it. All three carry a "−1," and they are related by the geometry, but they are three distinct operations:

- Spin is the ribbon's 2π spinor holonomy — the odd-n_C half-integer weight makes a 2π winding return −1 (Elie, grounded; SOLID). This is a genuine geometric mechanism.
- Charge sign is the sign of the SO(2)-weight — which way the winding runs (T2470, proved; SOLID). It is *not* a spinor holonomy. The v0.1 form q = $(-1)^n\cdot$(n/N_c) reproduces the four signed charges {0, −1/3, +2/3, −1}, but it does so because the winding-count assignment was *chosen* to match the weight-sign — it is a repackaging of the assignment, not an independent mechanism. The mechanism is T2470.
- Matter/antimatter is charge conjugation — reading the same oscillator from the other side of the membrane's $\mathbb{Z}_2$ (Section 3.5).

These three coincide numerically at −1 and all live on the same odd-dimensional, spin-type domain, which is why v0.1 was tempted to fuse them. But they are separate operators, and honesty requires saying so: spin is a holonomy, the charge sign is a weight-sign, matter/anti is a conjugation. The genuine unifying fact is weaker and still real — all three are native to D_IV^5 *because* it is odd-dimensional and a spin (type-IV) domain; none of them would exist on an even-dimensional or non-spin domain. That is the defensible referee sentence, not "one −1 does everything."

(The neutrino remains the minimal case: zero SO(2)-weight, so zero charge, plus the spinor half-twist — the bare Weyl spinor, the remainder.)

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5. The drive alphabet

If the commitment cycle sorts light into the section spaces of the compact dual $Q^5 = SO(7)/[SO(5)\times SO(2)]$, the loading counts are the low SO(7) representation dimensions — and, computed forward and target-innocent (Grace), they *are* the substrate integers:

Q^5 section space	dimension	equals
O(1) sections	7	g
spinor bundle	8	2^(N_c)
adjoint	21	N_c·g
O(2) sections	27	N_c^3

The 27 loadings are the quadratic functions on Q^5 — a finite drive alphabet, not a continuous knob. This resolves a Five-Absence concern: 27 is O(2) on our own geometry, not the E_6/Albert (forbidden GUT) 27. It also quantizes the "variable pump" that distinguishes up from down: the up and down loadings are different code words in a finite alphabet, so their relative phase (CP) is discrete and information-forced, not dialed. Whether {7, 8, 21, 27, …} forms a forced emission code with a physics-forced level is open (Section 7).

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6. Predictions and falsifiers

The framework is falsifiable at several points that do not depend on the open mass sector:

1. Charge: no fermion has |q| > 1; every charge is a multiple of 1/N_c = 1/3; exactly N_c + 1 magnitudes exist. Any exotic charge falsifies the winding count.
2. Spin: only spin-½ matter and integer-spin carriers; no spin-3/2 (no gravitino), no exotic-spin states. This is the same content as BST's standing no-SUSY-spectrum prediction, now geometric.
3. CPT exact; CP violated; matter dominant — required together, from one oscillator read on a slightly tilted membrane. A measured CPT violation falsifies the mechanism outright.
4. No-cloning, derived. Copying requires assembling from *committed* information, but committing an unknown superposition is measurement — it collapses. So the substrate can copy the sealed/encrypted state (the wrapper — allowed) but cannot commit one thing twice: the outcome exists only at the act of commitment, and the act is singular. A demonstrated perfect cloner of unknown accessible quantum states would falsify BST. (Consistent with the Yamaguchi–Kempf *et al.* encrypted-cloning demonstration, which clones the wrapper under a single-use key — exactly the boundary at commitment.)
5. Neutrino as minimal spinor — the lightest neutrino is the bare half-twist with zero SO(2)-weight (zero charge); the framework favors a nearly-massless ground state (normal ordering).
6. Mass predictions (the real falsifiers). The mass sector makes hard, target-innocent value predictions that a measurement can kill: s/d = (N_c+1)(N_c+2) = 20 (down-quark current-mass ratio); m_top = $v/\sqrt{2}$ = 174 GeV and m_charm = $\alpha \cdot v/\sqrt{2}$ = 1269 MeV (from v = m_p^2/(g·m_e), zero heavy-quark input); the Koide ratio = rank/N_c = 2/3 forcing m_μ/m_e = 206.9 from m_τ/m_e. Any of these moving outside its stated precision under improved measurement falsifies the corresponding mechanism. Per BST's falsifier program, these SM-observable predictions — not tabletop tests, which reproduce standard physics — are where the theory is actually at risk.

7. The honest ledger

This is the load-bearing section. Nothing outside this table should be read as a stronger claim than the table permits.

DERIVED / SOLID (proved theorem, target-innocent, or standard mathematics): - Charge — magnitude and sign — is the SO(2)-weight operator (T2470, proved). Quantized in units of 1/N_c, exactly N_c+1 magnitudes {0, 1/3, 2/3, 1}, capped at 1, no exotics. Five-Absence PASS. (Section 3.3) *Upgraded from v0.1's winding-holonomy story to a proved footing.* - Spin = ribbon half-twist ⇒ spin-½ + Fermi statistics; the spinor is native to the type-IV (spin-factor) domain; the 2π holonomy is the odd-n_C half-integer weight (Elie, grounded). Standard ribbon/framing mathematics. (Section 3.4) - CPT exactness as a structural consequence of "one oscillator, two readings." (Section 3.5) - The two posts = rank-2 torus; the poles = Shilov boundary $(S^4 \times S^1)/\mathbb{Z}_2$ — source-pinned to the polydisk theorem (Elie), not asserted. (Section 4, Appendix) - The drive alphabet {7, 8, 21, 27} = SO(7) rep dimensions = the substrate integers, computed forward. (Section 5) - No-cloning, derived from the commitment-collapse mechanism (copy the wrapper, commit once). (Section 6) - DOWN-QUARK current-mass ratios — BANKED (F506). d:s:b = 1:20:840 via the FK generalized Pochhammer (ν)_λ at ν = N_c; s/d = (N_c+1)(N_c+2) = 20 forced (0.5% vs current), zero parameters, target-innocent; single-row forced by the Q^5 ring $\mathbb{Z}[h]/h^6$. (Section 3.2) - The gen-1 inversion m_u < m_d is forced — the up ground (n=0, the doublet-flip of the down) sits below the down ground (degree 1). A structural result (proton stability), independent of the up's soft value. (Section 3.2)

PREDICTED (from forced quantities, target-innocent, magnitude at the stated precision): - m_top = v/$\sqrt{2}$ = 174 GeV (0.78%) — with v = m_p^2/(g·m_e) = 246.1 GeV forced independent of the top (F509); y_t = 1 by Cauchy–Schwarz. (Section 3.2) - m_charm = α·v/$\sqrt{2}$ = 1269 MeV (0.05%) — the charm Yukawa is α = 1/N_max, the boundary coupling one shell in. (Section 3.2)

NEAR-FORCED / CONDITIONAL (derivation-structure in hand, confirmed target-innocently, one clean gate remaining): - Charged leptons closed by Koide = rank/N_c, DERIVED from rank-2 (K672). Koide = the 45° tilt of the $\sqrt{}$mass vector; forced by N_gen = rank+1 (F86) and A^2 = rank (colorless filling, confirmed to 0.02%); self-carrying falsifier passes (only colorless obeys). With m_τ/m_e = 49·71 the muon follows at 0.05%. One gate: derive A^2 = rank from the Bergman overlap at the three F86 strata → then fully forced. (Section 3.2)

SYNTHESIS (the absolute-scale ledger, F511): from one dimensionful input, m_Planck → m_e (0.03%) → m_p → v = m_p^2/(g·m_e) = 246.1 GeV (0.04%), 8 of 9 charged fermion masses follow — 5 sub-percent (e, μ, τ, top, charm), 3 few-percent (d/s/b via the F506 ratios, absolute anchor ~5%), 1 soft (up n=0). The charged spectrum hangs off m_e and v with no second per-sector dimensionful anchor. (Not "all twelve": the neutrinos are open, the up is soft, the down absolute anchor is the loosest charged piece.)

CANDIDATE / SOURCED-BUT-UNBUILT (structure in hand, magnitude untested): - CP mechanism — the up-down twist — is now SOURCED (not merely candidate): the weak-isospin doublet is the Pin(2) doublet on the non-orientable Möbius locus (T1949, T2138, proved); the relative Pin(2) twist is the kink, and the same non-orientability gives parity violation — one locus, both. The *existence* of CP is grounded; only the *magnitude* (J) is untested (see Open). (Sections 3.5, 4) - The particle-to-count assignment (why ν, d, u, e sit at their SO(2)-weights) — a reading of the proved weight, awaiting a first-principles ordering. (Section 3.3)

OPEN (mass sector — the tower's three named lanes at EOD 2026-07-11): - The deep-bulk overlap — the up (n=0, ~22% soft) and down (n=1, ~5% soft) absolute masses are the two deepest quarks and the two softest values; fixing them is one computation (a proper Bergman-normalized overlap at the near-constant modes), not two problems. (Casey's two-address reading of the up was tested against the neutron–proton splitting and leans no at ~1.5σ; revives if lattice (M_n−M_p)^QCD → ~3.0.) (Section 3.2) - The last Koide gate — now reduced (Elie) to a Schur condition: λ_singlet = λ_traceless (the Higgs couples equally to the two generation irreps, singlet ⊕ rank-dim traceless). Confirmed by data to 0.02%; the bank is computing both couplings from the Bergman overlap at the F86 strata → flips the leptons near-forced → forced. (Section 3.2, K672) - The neutrino sector — the one untouched sector; characterized as zero-winding → likely $m_1 \approx 0$, not a Koide triple (consistent with the banked normal ordering). Koide closes the *charged* leptons only. - The up-type interior (the charm's generation degree) via the boundary-cohomology. *The Pin(2)/Möbius doublet-flip of the down's {1,3,5} is CLOSED as disfavored* — the up reads its masses through the boundary, not a flip of the down's formula. (Section 3.2) - The c/u warm-cold step (gate-3 second half) — genuinely hard (not single-τ Boltzmann); held open, not fitted. - CP magnitude — does the sourced Pin(2)/Möbius kink land J ≈ 3×10^{-5} (reconciled with the triangle-area reading). The build. (Sections 3.5, 4) - The forced half-twist — why the geometry makes matter spin-½ rather than us noting it fits. (Section 3.4) - η_B magnitude — only the existence and sign of the asymmetry are the cant; the size is a separate quantity (a current 2σ lead, $2\alpha^4/3\pi$). (Section 3.5)

RETRACTED (v0.2): - The "one −1 does all three" unification (v0.1 Section 4). Corrected to three distinct operations (Section 4). - The $(-1)^n$ charge-sign holonomy as a *mechanism* — it is a repackaging of the assignment; the mechanism is T2470's weight-sign.

RELATED, BANKED ELSEWHERE (the static / Phase-1 side): the cosmological cluster — Ω_Λ = $c_3/(c_3+\chi)$ = 13/19, Ω_DM/Ω_b = rank^4/N_c = 16/3, n_s = 1 − n_C/N_max — three parameters on three Chern channels of the same Q^5 spectrum whose odd cohomology are these generations. (Reported separately; the static reading of the same manifold.)

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8. The construction that decides

Two constructions were at stake. One is now decided; the other is set up and buildable.

Masses — mechanized (the v0.2 "miss" was a wrong-target error). The generation states were built on D_IV^5 with the correct type-IV norm; against *current* masses (the commitment/boundary ruling) and read through the FK generalized Pochhammer, the down ratios bank (s/d = 20 forced, 0.5%), the up-type endpoints force with the top and charm masses predicted, and the leptons close on Koide = rank/N_c. What remains is to *force* the two identified pieces (Koide's geometric origin, the up-type interior via the doublet-flip cohomology) and close the neutrinos — three clean forcing lanes, not a structural wall.

CP — set up, not yet run. The up-down structure is now sourced (the Pin(2) doublet on the non-orientable Möbius locus, T1949/T2138 proved), so the kink is buildable. Two independent handles must agree — the Pin(2)-twist size and the triangle-area reading — and both must land J ≈ 3×10^{-5}. The *existence* of the twist is forced by the locus's non-orientability (the same source as parity violation); the *magnitude* is the remaining test. The disciplines are fixed in advance: the locus and its winding numbers pinned to source (FK/Hua), no fitting.

The pattern is the framework behaving as an honest theory: the spine (charge via T2470, spin, CPT, the source-pinned geometry) hardened; the mass sector, once named a miss, dissolved that miss on a corrected target (current not constituent) and rebuilt into two forced mechanisms with two masses predicted; CP awaits one number against a sourced structure. Nothing is being tuned — each branch is being made to derive, or its residual named for what it is (the up n=0 value soft; the two identified pieces awaiting a first-principles force).

Appendix: notation and the five integers

rank = 2, N_c = 3, n_C = 5, C_2 = 6, g = 7; N_max = N_c^3·n_C + rank = 137. Domain D_IV^5 = $SO_0(5,2)/[SO(5)\times SO(2)]$, rank 2, complex dimension 5, Shilov boundary $(S^4 \times S^1)/\mathbb{Z}_2$, compact dual Q^5 = $SO(7)/[SO(5)\times SO(2)]$. Bergman kernel diverges toward the Shilov boundary (the "poles"). Type IV = the spin-factor Jordan/Clifford domain (the spinor is native).

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v0.4, 2026-07-11 PM. Drafted by Keeper as the honestly-tiered framework record; built with Casey (the mechanism, the current-vs-constituent target ruling, the doublet-flip forcing of the up ground), Lyra (the type-IV norm, F506/F509/F510/F511, the forced electroweak scale, the absolute-scale map, the weight-assignment discipline), Elie (the blind degrees, the FK single-row topology, m_charm = $\alpha\cdot v/\sqrt{2}$, and the Koide = rank-2 derivation via θ=45°), Grace (the drive alphabet, the exp(−distance) spine, the two-address discriminator test, and the theorem-graph *restoration). v0.3 dissolved the v0.2 "structural miss" (a wrong-target error) and rebuilt the mass sector into two mechanisms; v0.4 folds in the two afternoon closes — Koide derived from rank-2 (near-forced, K672) and the absolute-scale map (8 of 9 charged masses from one input, F511). This is a framework, not a completed derivation; Section 7 is the honest ledger and Section 8 is what remains. The remaining forcing lanes — the last Koide gate (A^2=rank), the up-type interior via boundary-cohomology, and the neutrinos — are the way to a fully-forced tower.*